

You have been supplied with the following source files:

**TreeData.txt**

Open the evidence document, **evidence.doc**

Make sure that your name, centre number and candidate number will appear on every page of this document. This document must contain your answers to each question.

Save this evidence document in your work area as:

**evidence\_** followed by your centre number\_candidate number, for example: **evidence\_zz999\_9999**

A class declaration can be used to declare a record. If the programming language used does **not** support arrays, a list can be used instead.

One source file is used to answer **Question 3**. The file is called **TreeData.txt**

**1** A program stores data about birds using Object-Oriented Programming (OOP).

The class `Bird` stores the data about the birds:

| <b>Bird</b>                         |                                                                                                                                                                   |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>DistancePerHour : Real</code> | stores the number of kilometres per hour (km/h) the bird can fly (between 0.0 and 100.0 inclusive)                                                                |
| <code>Species : String</code>       | stores the species of the bird, for example Pigeon                                                                                                                |
| <code>XPosition : Real</code>       | stores the current horizontal position of the bird                                                                                                                |
| <code>YPosition : Real</code>       | stores the current vertical position of the bird                                                                                                                  |
| <code>Constructor()</code>          | initialises <code>Species</code> and <code>DistancePerHour</code> to the parameter values; initialises <code>XPosition</code> and <code>YPosition</code> to 500.0 |
| <code>GetPosition()</code>          | returns a string message that contains the horizontal and vertical position of the bird                                                                           |
| <code>GetSpecies()</code>           | returns the species of the bird                                                                                                                                   |
| <code>Move()</code>                 | takes a direction and number of minutes flying (between 0 and 500 inclusive) as parameters and updates the appropriate horizontal or vertical position            |